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M. Tech. (First Semester) Examination, Nov.-Dec. 2019

(Old Scheme)

(Elect. Engg. Branch)

(Specialization : High Voltage Engineering)

POWER SYSTEM PROTECTION

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 40

Note *: All question are compulsory. Attempt any* ***two*** *parts from (a), (b), (c). All questions carries 10 marks.*

Unit-I

1. (a) Explain the basic difference between a $C.T_s$ used for insturmentation and $C.T_s$ used for protection.

(b) What do you understand by a zone of protection? Discuss various zones of protection for modern power system.

(c) Explain the nature and causes of faults. Discuss the concequences of faults on a power system.

Unit-II

2. (a) Show that the amplitude and phase comperators are dual to each other. Describe any one electromagnetic phase comparator.

(b) Discuss the different types of distance relay characteristics. How are these characteristics derived in an electronmagnetic comparetor?

(c) Describe the operating principle constructional features and area of applications of reverse power or directional relay. What is a directional over current relay.

Unit-III

3. (a) What types of Protection scheme is employed for the Protection of a large power transformer against

short circuits? with neat sketches discuss its working principles.

(b) Discuss the protection employed against loss of excitation of an alternator.

(c) An 11KV, 100 MVA alternator is provided with different protection. The percentage of winding to be protected against phase to ground fault is 85%. The relay is set to operate when there is 20% out of balance current. Determine the value of the resistance to be placed in the neutral to ground connection.

Unit-IV

4. (a) Explain characteristics and functional diagrams of numerical relay.

(b) Discuss microprocessor and DSP based relays.

(c) Discuss integrated and multifunction protection schemes.

Unit-V

5. (a) Discuss the performance of circuit breuher when capacitive currents are interupted.

(b) What are the different tests carried out to prove the ability to circuit breaker.

(c) Calculate the RRRV of a 220 kV circuit breaker with earthed neutral. The short circuit test data obtained is as follows : https://www.csvtuonline.com

The current broken is symmetrical and the restrihing voltage has an oscillatory frequency of 15 kHz. The p.f. of the fault is 0·2.

Assume the S.C. to be an earthed fault.